

Context

Streamist is a streaming company that streams web series and movies for a worldwide audience. Every content on their portal is rated by the viewers, and the portal also provides other information for the content like the number of people who have watched it, the number of people who want to watch it, the number of episodes, duration of an episode, etc.

They are currently focusing on the anime available in their portal, and want to identify the most important factors involved in rating an anime. You as a data scientist at Streamist are tasked with identifying the important factors and building a predictive model to predict the rating on an anime.

Objective

To analyze the data and build a linear regression model to predict the ratings of anime.

Key Questions

1. What are the key factors influencing the rating of an anime?
2. Is there a good predictive model for the rating of an anime? What does the performance assessment look like for such a model?

Data Information

Each record in the database provides a description of an anime. A detailed data dictionary can be found below.

Data Dictionary

- title - the title of anime
- description - the synopsis of the plot
- mediaType - format of publication
- eps - number of episodes (movies are considered 1 episode)
- duration - duration of an episode in minutes
- ongoing - whether it is ongoing
- sznOfRelease - the season of release (Winter, Spring, Fall)

- years_running - number of years the anime ran/is running
- studio_primary - primary studio of production
- studios_colab - whether there was a collaboration between studios to produce the anime
- contentWarn - whether anime has a content warning
- watched - number of users that completed it
- watching - number of users that are watching it
- wantWatch - number of users that want to watch it
- dropped - number of users that dropped it before completion
- rating - average user rating
- votes - number of votes that contribute to rating
- tag_Based_on_a_Manga - whether the anime is based on a manga
- tag_Comedy - whether the anime is of Comedy genre
- tag_Action - whether the anime is of Action genre
- tag_Fantasy - whether the anime is of Fantasy genre
- tag_Sci_Fi - whether the anime is of Sci-Fi genre
- tag_Shounen - whether the anime has a tag Shounen
- tag_Original_Work - whether the anime is an original work
- tag_Non_Human_Protagonists - whether the anime has any non-human protagonists
- tag_Drama - whether the anime is of Drama genre
- tag_Adventure - whether the anime is of Adventure genre
- tag_Family_Friendly - whether the anime is family-friendly
- tag_Short_Episodes - whether the anime has short episodes
- tag_School_Life - whether the anime is regarding school life
- tag_Romance - whether the anime is of Romance genre
- tag_Shorts - whether the anime has a tag Shorts
- tag_Slice_of_Life - whether the anime has a tag Slice of Life
- tag_Seinen - whether the anime has a tag Seinen
- tag_Supernatural - whether the anime has a tag Supernatural
- tag_Magic - whether the anime has a tag Magic
- tag_Animal_Protagonists - whether the anime has animal protagonists
- tag_Ecchi - whether the anime has a tag Ecchi
- tag_Mecha - whether the anime has a tag Mecha
- tag_Based_on_a_Light_Novel - whether the anime is based on a light novel
- tag_CG_Animation - whether the anime has a tag CG Animation
- tag_Superpowers - whether the anime has a tag Superpowers
- tag_Others - whether the anime has other tags
- tag_is_missing - whether tag is missing or not

Learning Outcomes

- Exploratory Data Analysis
- Preparing the data to train a model
- Training a regression model
- Model evaluation
- Forward Feature Selection